**ARTIFICIAL INTELLIGENCE ASSESSMENT – 2**

**REPORT**

**Course Code: CSA1704 – Artificial Intelligence**

**Submitted By**

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**Acknowledgement**

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**Question 1 – Hospital Doctor Scheduling using Backtracking Search**

**Problem Statement**

A hospital needs to assign three doctors (D1, D2, and D3) to three work shifts namely Morning, Afternoon, and Night. Each doctor must be be assigned exactly one shift while satisfying the given scheduling constraints. The objective is to determine a valid schedule using the Backtracking Search Algorithm. This problem demonstrates the application of Constraint Satisfaction Problems (CSP) in Artificial Intelligence for solving scheduling and resource allocation problems efficiently.

**Solution**

The Hospital Doctor Scheduling problem is solved using the Backtracking Search algorithm. The algorithm assigns shifts to doctors one by one while checking all the given constraints. Whenever a constraint is violated, the algorithm returns to the previous assignment and tries another possible shift. This process continues until a valid schedule satisfying all the constraints is obtained.

**State Representation**

State = (D1, D2, D3)

Initial State = (-, -, -)

Goal State = (Morning, Afternoon, Night)

**Constraints**

• D1 cannot work in the Night shift.

• D2 must be assigned before D3.

• D3 cannot work in the Morning shift.

• Only one doctor can be assigned to each shift.

**Architecture**

Start

↓

Assign Shift

↓

Check Constraints

↓

Constraint Satisfied

↓

Yes → Assign Next Doctor

↓

No → Backtrack

↓

Goal State

**Result**

The Backtracking Search algorithm successfully generates a valid doctor schedule by satisfying all the given constraints.

**Question 2 – Robot Navigation using Breadth First Search**

**Problem Statement**

A robot is placed in a 5 × 5 grid environment where it has to move from the Start position to the Goal position while avoiding obstacles. The robot can move only in four directions namely Up, Down, Left, and Right. Each movement has a cost of one unit. The objective is to determine the shortest path using the Breadth First Search (BFS) algorithm. This problem demonstrates the application of AI search techniques in robot navigation and path planning.

**Solution**

The Robot Navigation problem is solved using the Breadth First Search (BFS) algorithm. The algorithm starts from the initial position and explores all neighbouring nodes level by level using a queue. Since every movement has the same cost, BFS always finds the shortest path from the start node to the goal node.

**State Representation**

State = (Current Position)

Initial State = Start Node (S)

Goal State = Goal Node (G)

**Operators**

• Move Up

• Move Down

• Move Left

• Move Right

**Architecture**

Start Node

↓

Generate Successor Nodes

↓

Queue

↓

Expand Node

↓

Goal Test

↓

Goal Reached

**Result**

The Breadth First Search algorithm successfully identifies the shortest path from the start node to the goal node.

**Question 3 – Autonomous Rescue Robot using Online Search and Uniform Cost Search**

**Problem Statement**

An autonomous rescue robot is deployed in a disaster-affected area to locate survivors safely. The robot operates in a partially observable environment containing blocked paths and risky zones. It continuously senses the environment, updates its knowledge, and selects the least-cost path to reach the survivor. This problem demonstrates the use of Online Search Agents and Uniform Cost Search in intelligent decision-making.

**Solution**

The rescue robot is implemented as an Online Search Agent. It continuously collects information from the environment using sensors and updates its internal knowledge base. The Uniform Cost Search (UCS) algorithm is used to identify the least-cost path by expanding the node with the minimum cumulative cost first. This enables the robot to reach the survivor efficiently while minimizing the total travel cost.

**Environment**

• Safe Area

• Blocked Area

• Risk Zone

• Survivor Location

**Actions**

• Move Forward

• Turn Left

• Turn Right

• Sense Environment

• Update Knowledge

• Rescue Survivor

**Architecture**

Environment

↓

Sensors

↓

Knowledge Base

↓

Uniform Cost Search

↓

Decision Making

↓

Actuators

↓

Survivor Reached

**Result**

The Online Search Agent successfully reaches the survivor by selecting the minimum-cost path using the Uniform Cost Search algorithm.